



Indian Sign Language (ISL) Recognition

A Simple Explanation of the Problem, the Data and the AI Model (PRAICP-1000)

1. What Problem Is This Project Solving?

Sign language is how many deaf and hard-of-hearing people communicate every day. The problem is that most hearing people never learn it, so there is a real communication gap in schools, hospitals, banks and everyday public life.

This project builds an AI model that looks at a photograph of a single hand sign from the Indian Sign Language (ISL) alphabet and automatically tells you which letter it represents. The long-term idea is that a tool like this, built into a phone or web app, could act as a translator that helps someone who doesn't know sign language understand a person who is signing to them.

Why Only Some Letters?

Not every letter in this project can actually be shown in a single photo. Two letters — J and Z — are normally signed with a hand movement (a little drawn shape in the air), so a single still picture can't capture them. That's why this project works with 24 letters (A through Y, skipping J and Z), each of which is a fixed, still hand shape that a photo can fully capture.

2. The Data: Thousands of Hand Photos

The project used 4,972 real photographs of hands, already sorted by DataMites into 24 folders — one folder per letter. Each photo shows a single hand, making one sign, against a plain background.

- Training images: 3,480 — used to actually teach the model.
- Validation images: 746 — used while building the model, to check progress and pick the best settings.
- Test images: 746 — locked away and only looked at once, right at the very end, to give an honest final score.

Two Challenges in the Photos

First, some letters had a lot more photos than others — the letter B had 259 photos while the letter H had only 116, a little over twice as many. If ignored, the model could get lazy and favour the letters it saw more often. The project fixed this by making sure every split (training/validation/test) kept the same mix of letters, and by giving extra importance to the rarer letters during training.

Second, the photos came in three different sizes and shapes — some wide, some very large, some even taller than they are wide. Since a computer model needs every image to be exactly the same size to compare them properly, every photo was resized to a standard small square (128 by 128 pixels) before the model ever saw it.

3. How the AI Model Actually Reads a Hand Sign

The model used here is a Convolutional Neural Network (CNN), a type of AI that is especially good at looking at pictures. You can think of it as working in stages, a bit like how a person might describe a hand sign step by step:

- First it notices simple things — edges, curves, the outline of fingers.
- Then it combines those into bigger shapes — a curled finger, a straight thumb, a closed fist.
- Finally it compares the overall hand shape against everything it has learned, and picks the one letter that matches best, along with how confident it is.

To stop the model from simply "memorising" the training photos instead of really learning the hand shapes, the project showed it slightly altered copies of each training photo during learning — a bit rotated, a bit shifted, a bit zoomed — so it would get used to natural variation in how a person actually holds up their hand. One thing it deliberately avoided was flipping images like a mirror, because a mirrored left hand can turn into a sign that looks like a completely different (and wrong) right-hand letter.

Two Versions Were Built and Compared

A first, "baseline" version of the model was built using reasonable, sensible settings, just to have something to compare against. Then the project used a tool called Keras Tuner to automatically try out several different combinations of settings (such as how many filters to use, how much "forgetting" (dropout) to apply, and how fast the model should learn) and kept the combination that worked best. This improved, "tuned" version is the final model.

4. How Well Did It Actually Work?

Model	Validation Accuracy	Test Accuracy	Test F1-Score
Baseline CNN	99.46%	99.33%	0.993
Final Tuned CNN	99.87%	100.00%	1.000

Results on 746 unseen test photographs, across 24 sign letters.

The baseline model was already very accurate, correctly recognising about 99 out of every 100 test photos. The tuned model did even better, and on this particular test set it got every single photo right — all 746 of them, across all 24 letters.

A Quick, Honest Word About “100%”

A perfect score sounds almost too good, and it's worth being honest about what it does and doesn't mean. It shows the model learned this specific dataset extremely well — the photos were all taken in a fairly similar way (plain background, one hand, similar lighting), which makes the task more consistent than, say, random photos taken by hundreds of different people in messy real-world settings. So a perfect score here is strong evidence the model is excellent on this kind of photo, but it is not a guarantee it would still be 100% correct on a brand-new person's hand, photographed with a different phone, background, or lighting. That kind of real-world testing is the natural next step.

5. What This Means in Practice

- This model could become the first building block of an app that reads a photo of a hand sign and instantly shows the matching letter — useful in schools, service counters, or simple everyday conversations.
- It currently only recognises single, still letters (no words, no sentences, and no signs that need movement, like J or Z) — real conversations would need a more advanced version that understands video and full words.
- Before any real-world use, it should be tested on new photos from different people, cameras and backgrounds, to make sure the excellent lab results carry over to everyday conditions.
- Even a simple letter-recognition tool like this is a meaningful step towards making everyday communication more accessible for people who use sign language.

In short: using about 5,000 labelled hand photos and a well-tuned CNN, this project built a model that reads static Indian Sign Language letters with very high accuracy — a strong proof that this kind of accessibility tool is realistic to build, while also being honest that more real-world testing is the natural next step before it leaves the lab.